

NUTRIFY: Gym Management and Fitness Planning System

1. Project Overview

Nutrify is a comprehensive gym management system designed to streamline operations at fitness centers. It enables efficient handling of member registration, trainer assignments, personalized fitness and nutrition plans, and equipment monitoring.

The purpose of the system is to improve organization, automate manual processes, and enhance communication between gym members, trainers, and administrators.

Target Users: Fitness Enthusiast, Fitness Trainers, Gym Owner, Administrative Officer and Maintenance Officer

2. Project Objectives

General Objective: To develop an integrated system that manages gym operations and provides personalized fitness and nutrition planning for members.

Specific Objectives:

- Automate membership registration and management
- Allow trainers to create and manage personalized fitness plans
- Track gym equipment status and maintenance
- Provide real-time notifications and updates
- Generate reports for management decision-making

3. Project Scope

In Scope:

- User authentication
- Membership management
- Fitness and nutrition plan creation
- Trainer scheduling and assignment
- Equipment maintenance tracking
- Notification and reporting system

Out of Scope:

- Mobile app version
- Wearable device integration
- AI-based recommendations

4. Project Timeline

Week	Activities
Week 1	Planning, system design, requirements gathering
Week 2	Development of core modules
Week 3	Testing and debugging
Week 4	Deployment and documentation

5. Tools and Technologies

component	Technology Used
Frontend	HTML, CSS, JavaScript
Backend	PHP
Database	MySQL
Development Environment	XAMPP, Visual Studio Code

6. System Architecture

The system follows a three-tier architecture: client-side (user interface), server-side (business logic), and database layer (data storage).

7. Functional Requirements

Feature	Description
User Authentication	Login and account management
Membership Management	Handles registration and status
Fitness Plan	Trainer-created personalized plans
Trainer Management	Schedules and assignments
Equipment Maintenance	Tracks equipment status
Notification System	Sends alerts and updates

8. Non-Functional Requirements

- Usability: User-friendly interface
- Performance: Fast response time
- Security: Data protection and authentication
- Reliability: Minimal downtime

9. Risk Management Plan

Risk Type	Description	Impact
Technical	System bugs	High

Schedule	Development delays	Medium
Resource	Limited tools/manpower	Medium

10. Testing Plan

Test Type	Description
Unit Testing	Test individual components
Integration Testing	Check module interaction
System Testing	Full system testing
User Testing	Feedback from users

11. Deployment Plan

Install required software, set up database, configure server, deploy system, and train users.

12. Maintenance Plan

- Updates
- Bug fixing
- Monitoring

- Backup

13. Conclusion

Nutrify improves gym efficiency by automating processes and enhancing communication. It ensures better service, organization, and user experience.